

KESHAV KRISHNA

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Product engineer with three years shipping ML and NLP systems to production, including consumer systems serving 100M+ users. Currently building risk analytics tooling at Moore Capital Management after extending a full-time summer internship into a part-time fall role. I own features from an ambiguous conversation through production and measure model quality with explicit evaluations.

EDUCATION

New York University, Courant Institute — MS, Computer Science <i>GPA 3.95/4.0; Machine Learning, Building LLM Reasoners, Honors Algorithms</i>	Aug 2025 – May 2027 New York City
Indian Institute of Technology, Ropar — BTech, Computer Science and Engineering <i>GPA 8.74/10; Merit Scholarship for top academic performance</i>	Aug 2019 – Jun 2023 Ropar, India

EXPERIENCE

Moore Capital Management — Risk Technology Intern <i>Risk analytics tooling and dashboards for internal users; full-time summer, part-time fall</i> – Building internal risk dashboards and reporting tools used by risk and operations stakeholders; own features end to end, from a verbal request to something in production days later. – Python data workflows feeding those surfaces, with the interface layer in HTML/CSS; iterating directly on feedback from the people who use the tools daily rather than through a spec handoff.	May 2026 – Present New York City
JioSaavn — Software Engineer, Data Science <i>Production ML and NLP in a consumer app serving 100M+ users</i> – Built recommendation and personalization systems on embeddings and vector retrieval (Spark, Hive, Redis, Qdrant), driving 1M impressions, 800K clicks, and 500K streams per day in production. – Shipped a LLaMA3-8B-Instruct conversational playlist system doing intent understanding and retrieval-driven song selection—an LLM feature in front of real users, deployed on Docker and Kubernetes, with the latency, cost, and failure-mode work that implies. – Ran a model bake-off across SetFit, HingRoBERTa, mURIL, and LLaMA3-8B-Instruct for multilingual playlist moderation, reaching 85% accuracy against a manually labeled evaluation set. – Worked across product, editorial, and infrastructure teams in a small data-science group where the same person wrote the model, the pipeline, and the deployment.	Jul 2023 – Jul 2025 Mumbai, India
GE Healthcare — Software Intern, Medical Computer Vision <i>Detection and de-identification pipelines for clinical imaging</i> – Built YOLO-based medical-image de-identification and RGB/IR people-detection pipelines over 20K+ images.	May 2022 – Jul 2022 Bangalore, India

SELECTED TECHNICAL PROJECTS

FlashAttention-2 — from-scratch Triton implementation – Forward and backward kernels with online softmax, tiling, causal masking, and activation recomputation; cut attention memory from $O(N^2)$ to $O(N)$, enabling 65K-token sequences. – Profiled Transformer models up to 2.7B parameters with NVIDIA Nsight Systems and PyTorch memory snapshots, benchmarked against vanilla PyTorch and <code>torch.compile</code> baselines.	2026
Transformer language model from scratch – 17M-parameter model in PyTorch built from low-level components (BPE tokenization, RoPE, RMSNorm, SwiGLU, causal attention, AdamW, cosine LR scheduling), with systematic ablations across pre- vs. post-norm, RoPE vs. NoPE, and SwiGLU vs. SiLU.	2026

RESEARCH AND PUBLICATIONS

Phase-Adaptive Generation: Efficient Decoding for Diffusion Language Models

Feb 2026 – May 2026

- NYU project under Prof. Greg Durrett. Analyzed generation dynamics of Dream/LLaDA-style diffusion LMs via per-token refinement traces, revealing interpretable phase structure: template spans stabilize immediately while arithmetic and answer-finalization spans demand substantially more refinement.
- Trained Transformer predictors over 138K block-level control tuples to forecast per-block refinement budgets; integrated into the LLaDA decoding loop, matching AdaBlock's 89.5% GSM8K accuracy with approximately 21% fewer forward evaluations. [Report]

Alignment and Reasoning RL for Mathematical LLMs

Spring 2026

- End-to-end post-training pipeline: zero-shot evaluation, SFT, GRPO with verified rewards, masked losses, entropy tracking, and policy-gradient updates. Fine-tuned Qwen2.5-Math-1.5B on chain-of-thought data, improving MATH accuracy from 56.4% to 60.4%.

ML-Driven Program Phase Prediction for Dynamic Hardware Resource Optimization

Jan 2026 – May 2026

- Independent study with Prof. Mohamed Zahran, NYU Courant. Unsupervised clustering over temporal traces of multicore workloads defines interpretable pressure states, distilled into compact predictors with sub-KB model state.

Publications

2024 – 2025

- *Advancements in Cache Management: A Review of Machine Learning Innovations* — Frontiers in AI, 2025. Single-author, 10+ citations; reviewed 40+ papers on ML for cache management.
- *LLC Intra-set Write Balancing for Non-Volatile Memory Caches* — arXiv, 2024. Bachelor's thesis research.

TECHNICAL SKILLS

Product / full-stack: React, JavaScript, HTML/CSS, FastAPI, Postgres, Redis, Docker, Kubernetes, Git, Linux

LLMs in production: Anthropic and OpenAI APIs, agents and tool use (MCP), evals and benchmarking, context management, RAG and vector search (Qdrant), SFT/GRPO/RLHF, PyTorch, Hugging Face, tokenizers, vLLM

Data / infrastructure: Python, C/C++, Spark, Hive, Airflow, Triton kernel development, NVIDIA Nsight Systems, Slurm/HPC, reproducible pipelines